Quiz 8 — Graphs — how the class did

9

students

Part 1 (Quiz 8A, Q1–Q3)

6.2 /10 mean

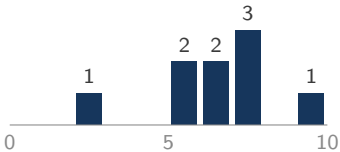
median 6.25

Part 2 (Quiz 8B, Q4–Q5)

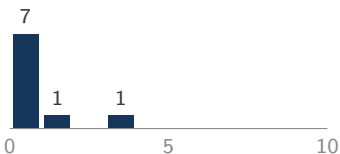
0.6 /10 mean

median 0

Part 1 (Quiz 8A): students per score out of 10



Part 2 (Quiz 8B): students per score out of 10



An external cost: social cost and the two quantities

SC = S + harm; the market stops where S meets D, the efficient quantity where SC meets D.

SC drawn (S + harm) (1 pt)	0%
market quantity (0.5 pt)	0%
efficient quantity (1 pt)	0%

bars: the class's share of the points of each item · task overall 0 %

Answers: version A: market quantity 4, efficient quantity 2 · version B: market quantity 3, efficient quantity 1

Most often: Not reached (one attempt drew a steeper line). SC = S + €4: a line PARALLEL to S, €4 higher; the efficient quantity is where SC meets D.

A tariff: revenue, DWL, CS and PS

The areas with the tariff, as numbers and labels.

tariff revenue T (0.5 pt)	0%
DWL (0.5 pt)	0%
CS (0.5 pt)	0%
PS (0.5 pt)	0%
labels T and DWL (0.25 pt)	0%
labels CS and PS (0.25 pt)	0%

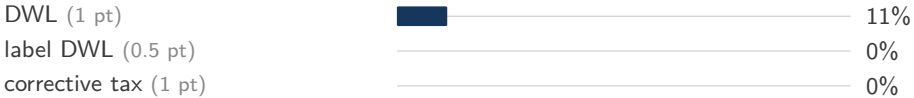
bars: the class's share of the points of each item · task overall 0 %

Answers: version A: tariff revenue T 2, DWL 1, CS 12.5, PS 4.5 · version B: tariff revenue T 2, DWL 1, CS 8, PS 2

Most often: Not reached by anyone. T = €1 × imports with the tariff; DWL = the two small triangles beside T; CS above the tariff price, PS below it.

An external cost: DWL and the corrective tax

DWL between SC and D from the efficient to the market quantity; the tax equal to the harm per unit.



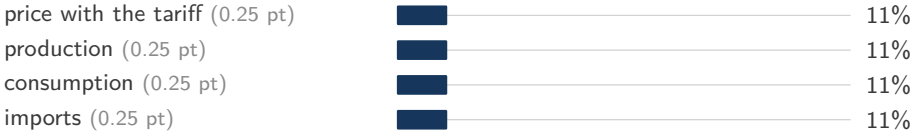
bars: the class's share of the points of each item · task overall 4 %

Answers: version A: DWL 4, corrective tax 4 · version B: DWL 4, corrective tax 4

Most often: Not reached. DWL $= \frac{1}{2} \times €4 \times$ (market − efficient quantity) $= 4$; the corrective tax equals the harm, €4 per pot.

A tariff: price and quantities

Price = world price + tariff; production, consumption and imports there.



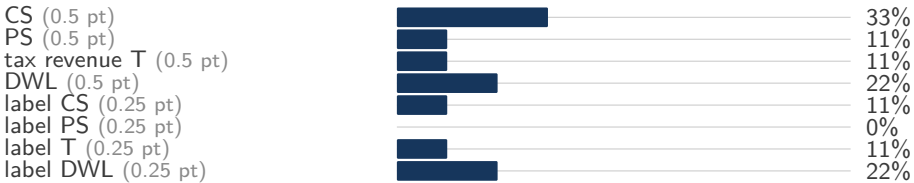
bars: the class's share of the points of each item · task overall 11 %

Answers: version A: price with the tariff 4.5, production 3, consumption 5, imports 2 · version B: price with the tariff 3.5, production 2, consumption 4, imports 2

Most often: Almost no one reached it. The tariff price = world price + €1; read production on S and consumption on D at that price.

A tax on sellers: CS, PS, revenue and DWL

The four areas as numbers, and their labels on the graph.



bars: the class's share of the points of each item · task overall 17 %

Answers: version A: CS 4.5, PS 4.5, tax revenue T 6, DWL 1 · version B: CS 2, PS 2, tax revenue T 4, DWL 1

Most often: About half left the areas blank; the letters CS, PS, T and DWL were rarely written on the graph. CS: under D above the buyers' price; PS: above S below the sellers' price; T = tax × quantity; DWL: the small triangle out to the old quantity.

Opening to trade

At the world price: production on S, consumption on D, imports = the gap.



bars: the class's share of the points of each item · task overall 22 %

Answers: version A: production 2, consumption 6, imports 4 · version B: production 1, consumption 5, imports 4

Most often: Two thirds of the class did not reach this question. At the world price: production where it meets S, consumption where it meets D, imports = the gap.

A tax on sellers: quantity and the two prices

S moves up by the tax; buyers pay on D at the new quantity, sellers keep that minus the tax.



bars: the class's share of the points of each item · task overall 65 %

Answers: version A: quantity sold 3, buyers pay 6.5, sellers keep 4.5 · version B: quantity sold 2, buyers pay 5.5, sellers keep 3.5

Most often: Half got the quantity: S + tax is S moved up by €2 everywhere (parallel), so it meets D at 3 thousand (version A) or 2 thousand (B). Sellers keep the buyers' price − €2.

A shift: 2 thousand more at every price

Move the curve 2 thousand to the right; E2 at the new crossing.



bars: the class's share of the points of each item · task overall 81 %

Answers: version A: price at E2 4, quantity at E2 4 · version B: price at E2 6, quantity at E2 5

Most often: Right for most. Lower costs move SUPPLY, a new bike lane moves DEMAND; 2 thousand more at every price = 2 squares to the right. Letter the new curve and E2.

Equilibrium from formulas

Draw Qd and Qs from their formulas; E1 where they cross.



bars: the class's share of the points of each item · task overall 86 %

Answers: version A: price at E1 5, quantity at E1 3 · version B: price at E1 5, quantity at E1 4

Most often: Right for most. Solve Qd = Qs first ($8 - P = P - 2$ gives $P = 5$), then mark E1 at that crossing.

Equilibrium from a table

Draw D and S through the table's points; read P* and Q* at the crossing.



bars: the class's share of the points of each item · task overall 91 %

Answers: version A: P* 5, Q* 5 · version B: P* 6, Q* 4

Most often: P* and Q* right for everyone. The crossing was found, but 6 of 9 did not write the letter E at it (a dot or dashed lines only).